

Boundary Completion and Cycle Memory in Zero Boundary Algebra

A rigorous 3–6–8 construction from guarded transitions,
policy equivariance, and auditable histories

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Abstract

The expression “3, 6, 8 rather than 3, 6, 9” is mathematically meaningful only after the counted objects and permitted operations are specified. This article develops such a specification from the three directional boundary phases of Zero Boundary Algebra (ZBA). On an explicitly defined phase-only linear model, three guarded transitions generate the full nine-dimensional matrix algebra $M_3(\mathbb{C})$; separating its scalar identity direction yields the eight-dimensional Lie algebra $\mathfrak{sl}_3(\mathbb{C})$. A general completion theorem distinguishes the linear spans of reversible relabellings, relabellings with total resets, and domain-sensitive operations, giving dimensions 5, 7, and 9 in the three-state case, and corresponding traceless dimensions 4, 6, and 8.

Endpoint matrices, however, erase repeated cycles. Introducing a formal crossing variable z gives a faithful representation of the directed three-cycle path algebra as polynomial matrices with lower-triangular constant term. Its trace-zero part is a standard polynomial Iwahori subalgebra; setting $z = 1$ recovers the eight-dimensional endpoint algebra, while formally inverting z produces the loop algebra of $\mathfrak{sl}_3$. The construction therefore identifies exactly where cycle information is discarded, rather than confusing endpoint symmetry with complete provenance. Two further results characterize when policy-dependent reset commutes with polarity reversal and quantify the exponential collapse of distinct mirror/reset histories onto one structural transformation. A reset-stable but mirror-sensitive policy supplies an explicit counterexample to unconditional commutation.

The proposed contribution is this ZBA-specific formalization, loss analysis, policy correction, and reproducible test package. Matrix, path, Iwahori, and Lie algebras themselves are established mathematics. Exact finite checks support the constructions; no new physical law, cryptographic primitive, or measured deployment advantage is asserted.

Keywords: typed state calculus; guarded transition; path algebra; Iwahori algebra; provenance; reset; equivariance; $\mathfrak{sl}_3$; information loss.

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1 Problem, scope, and contribution

A short sequence of numerals does not determine a mathematical theory. An arithmetic progression with common difference three gives 3, 6, 9. A different question—which nontrivial directions remain after the scalar direction is removed from three-state linear transformations—gives eight. Neither answer invalidates the other. The research problem is to replace an evocative numerical contrast with an operational construction whose assumptions, invariants, and information losses can be checked.

ZBA 1.1 supplies a concrete starting point. Its public specification distinguishes three boundary tags, polarity, validation evidence, and audit history; its mirror changes polarity, whereas its boundary operations move between phases. The specification and companion verifier were inspected on GitHub and Hugging Face [1, 2, 3]. This article studies a deliberately restricted mathematical projection of that system, then states what must be added to recover information discarded by the projection.

The resulting thesis is not that nine is wrong. It is

$$\begin{array}{l} \text{three phases} \longrightarrow \text{nine endpoint directions} \\ \text{nine endpoint directions} = \text{one scalar} + \text{eight traceless directions.} \end{array} \quad (1)$$

A second thesis is equally important: a completed cycle may restore a phase without restoring the past. Retaining phase history changes the algebra from a finite endpoint algebra to an infinite-dimensional path algebra.

The article contributes four linked constructions relative to the inspected ZBA specification. First, it derives an explicit endpoint matrix algebra from phase guards, and a general dimension ladder measuring the difference between total transformations and partial-domain observations. Second, it constructs an exact cycle-retaining representation and identifies its specialization and localization. Third, it supplies a necessary-and-sufficient policy condition for mirror/reset commutation. Fourth, it proves an audit-history collapse bound and turns these distinctions into executable tests and a falsifiable engineering programme.

These are not claims of having discovered $SU(3)$, matrix units, path algebras, or Iwahori algebras. Those subjects have substantial prior literature [4, 5, 6, 8]. The proposed research object is their explicit use to sharpen ZBA: which operations generate which algebra; what histories a representation identifies; and which policy assumptions are necessary for its advertised equalities.

2 Typed foundations and a necessary separation of mirrors

2.1 The phase-only projection

Let

$$P = \{-1, 0, +1\}, \quad B = \{b_0, b_1, b_2\},$$

where

$$b_0 = \text{boundary } (0), \quad b_1 = \text{emerging } (+0), \quad b_2 = \text{entering } (-0).$$

These are three distinct control tags, not three distinct real zeros. The scalar erasure map sends all of them to the real number zero. The phase-only successful cycle is

$$b_0 \xrightarrow{\text{Emerge}} b_1 \xrightarrow{\text{Enter}} b_2 \xrightarrow{\text{Seal}} b_0. \quad (2)$$

This is the ordinary cyclic ordering of the public specification, started at a different vertex. The genesis exception, rejection, recovery, asset identity, actor authorization, and validation

predicates are not encoded by this three-vertex cycle. Throughout the cycle-algebra sections, those additional guards are assumed satisfied for the paths under discussion. A matrix identity is not an authorization decision.

For structural-policy analysis we use reduced states

$$s = (p, b, d, e),$$

with polarity p , phase b , a nonnegative depth d , and evidence status

$$e \in \{\text{claimed, checked, sealed, rejected}\}.$$

Other ZBA fields are held fixed rather than presumed nonexistent.

2.2 Polarity reversal is not boundary reflection

The existing structural mirror is

$$M(p, b, d, e) = (-p, b, d, e).$$

It leaves b unchanged. Let C_B advance the boundary index by one modulo three. Then

$$MC_B = C_B M.$$

On $P \times B$, these two transformations generate $C_2 \times C_3$, a cyclic group of order six, not the nonabelian triangle symmetry group.

A *different*, optional operation is boundary reflection:

$$J(b_j) = b_{-j \bmod 3}.$$

Thus J fixes b_0 and swaps b_1, b_2 . Now

$$J^2 = \text{id}, \quad C_B^3 = \text{id}, \quad JC_B J = C_B^{-1},$$

and $\langle C_B, J \rangle \cong D_3 \cong S_3$. This is a comparison symmetry of the phase model. It does not identify J with ZBA's polarity mirror, nor authorize reversal of a recorded workflow.

The modular mnemonic is exact but likewise typed. In $\mathbb{Z}/9\mathbb{Z}$,

$$6 = -3, \quad 8 = -1, \quad 9 = 0,$$

so multiplication by eight exchanges three and six. Multipliers two and five also exchange that particular pair. Eight is distinguished by being the unique nonidentity multiplier satisfying $u^2 = 1$ modulo nine: from $9 \mid (u-1)(u+1)$, and the fact that the factors cannot both be divisible by three, one obtains $u = 1$ or $u = 8$ modulo nine. On $\{0, 3, 6\}$, translation by three and multiplication by eight represent the phase-cycle and phase-reflection construction after an explicitly chosen relabelling. That relabelling is not an identification of polarity with phase.

Interpretation. The number six can count group elements, vector-space dimensions, directed edges, or symmetric pairs. Equal counts do not make these objects identical. Keeping their types separate is essential to any defensible 3–6–8 claim.

3 Guarded boundary completion

Let $V = \mathbb{C}^3$ with basis e_0, e_1, e_2 corresponding to the three phases. Write E_{ij} for the matrix sending e_j to e_i and every other basis vector to zero. Then

$$E_{ij}E_{kl} = \delta_{jk}E_{il}. \quad (3)$$

Represent the three guarded arrows by

$$A_{\text{emerge}} = E_{10}, \quad A_{\text{enter}} = E_{21}, \quad A_{\text{seal}} = E_{02}. \quad (4)$$

A noncomposable product is zero. This zero is the zero vector or zero map—a failed branch in the linear representation—not the valid boundary state e_0 .

Theorem 3.1 (Three-phase endpoint completion). *The unital associative complex algebra generated by the three matrices in (4) is $M_3(\mathbb{C})$.*

Proof. The generators provide the three length-one arrows. Their composable length-two products give the other off-diagonal units:

$$E_{21}E_{10} = E_{20}, \quad E_{02}E_{21} = E_{01}, \quad E_{10}E_{02} = E_{12}.$$

Length-three cycles give the three diagonal units. Thus all nine E_{ij} belong to the generated algebra. Since these units form a basis of $M_3(\mathbb{C})$, the generated algebra is exactly $M_3(\mathbb{C})$. $\square$

Theorem 3.2 (One boundary filter suffices in every finite dimension). *For $n \geq 2$, let $Ce_j = e_{j+1 \bmod n}$ and $P_0 = E_{00}$ on $\mathbb{C}^n$. Then*

$$\text{Alg}_{\mathbb{C}}(C, P_0) = M_n(\mathbb{C}).$$

Proof. Since $C^{-1} = C^{n-1}$, all powers needed below are in the algebra. Direct application to each basis vector gives

$$C^i P_0 C^{-j} = E_{ij}, \quad 0 \leq i, j < n. \quad (5)$$

Hence every matrix unit is generated. $\square$

The filter P_0 asks whether the source state is the boundary state. It is different from a total reset, which sends *every* source basis state to the same target. This distinction, not a replacement of ordinary arithmetic, is the mechanism behind completion.

The proof is elementary matrix algebra; related general generation results belong to established representation theory [4, 5]. The ZBA-specific step is the choice of generators from guarded phase operations. Their complex linear span is an analytical envelope. An arbitrary complex linear combination of transitions need not itself be a permitted state transition, a stochastic update, or a physically realizable operation.

4 A dimension ladder: what resets cannot observe

The preceding theorem can be sharpened by comparing three different operator families. Let $\varepsilon = (1, \dots, 1)$ and $W = \ker \varepsilon \subset \mathbb{C}^n$. Let $\mathcal{P}_n$ be the complex span of all $n \times n$ permutation matrices. For every target i , define the constant reset

$$K_i = e_i \varepsilon.$$

Every column of K_i is e_i . Define

$$\mathcal{A}_\varepsilon = \{A \in M_n(\mathbb{C}) : \varepsilon A = \lambda \varepsilon \text{ for some } \lambda \in \mathbb{C}\}.$$

Equivalently, $\mathcal{A}_\varepsilon$ consists of operators preserving the hyperplane W .

Theorem 4.1 (Permutation–reset–guard dimension ladder). *For $n \geq 2$,*

$$\dim \mathcal{P}_n = (n-1)^2 + 1, \quad (6)$$

$$\text{span}(\mathcal{P}_n, K_0, \dots, K_{n-1}) = \mathcal{A}_\varepsilon, \quad \dim \mathcal{A}_\varepsilon = n^2 - n + 1, \quad (7)$$

$$\text{Alg}_{\mathbb{C}}(\mathcal{A}_\varepsilon, P_0) = M_n(\mathbb{C}), \quad \dim M_n(\mathbb{C}) = n^2. \quad (8)$$

Proof. Every linear combination of permutation matrices has all row sums and all column sums equal to one common scalar. The space of matrices with zero row and column sums has dimension $(n-1)^2$: the upper-left $(n-1) \times (n-1)$ block determines the remaining entries. It is spanned by

$$(e_i - e_{n-1})(e_j - e_{n-1})^T, \quad i, j < n-1.$$

Each such matrix is a difference of two permutation matrices agreeing off the selected two rows and columns. Adding the identity gives (6).

The reset span consists of matrices $v\varepsilon$ and has dimension n . Its intersection with $\mathcal{P}_n$ is one-dimensional: constant row sums require all coordinates of v to coincide. Consequently the combined span has dimension

$$(n-1)^2 + 1 + n - 1 = n^2 - n + 1.$$

All its generators satisfy $\varepsilon A = \lambda \varepsilon$. This latter condition imposes $n-1$ independent linear constraints on $M_n(\mathbb{C})$, so $\dim \mathcal{A}_\varepsilon = n^2 - n + 1$ and the two spaces agree. The set $\mathcal{A}_\varepsilon$ is an algebra because the column-sum scalars multiply under composition. Finally it contains the cyclic permutation C , so Theorem 3.2 proves (8). $\square$

For three phases the associative dimensions are

$$\boxed{5 \longrightarrow 7 \longrightarrow 9.} \quad (9)$$

All three spaces contain I , whose trace is nonzero. Removing the scalar direction gives traceless dimensions

$$\boxed{4 \longrightarrow 6 \longrightarrow 8.} \quad (10)$$

In particular, the six elements of S_3 span only five matrix directions in its natural three-state permutation representation. Counting transformations and counting linearly independent directions are different operations.

4.1 The exact information supplied by a guard

Consider the linear map

$$\Phi : M_n(\mathbb{C}) \longrightarrow W^*, \quad \Phi(A) = \varepsilon A|_W.$$

Its kernel is $\mathcal{A}_\varepsilon$ and it is onto. To see surjectivity, extend any functional on W to a row vector f on $\mathbb{C}^n$ and choose A with $\varepsilon A = f$. Therefore

$$M_n(\mathbb{C})/\mathcal{A}_\varepsilon \cong W^*$$

as *vector spaces*. This is not asserted to be a quotient of associative algebras: $\mathcal{A}_\varepsilon$ is generally not an ideal.

The missing $n-1$ dimensions are source-sensitive aggregate responses. Total deterministic transformations cannot expose them through their column sums because each source sends unit mass somewhere. A domain filter can respond differently to different sources. For three states, exactly two such directions are missing from the reset algebra.

4.2 Why six also appears in ordinary three-state rate models

With column-state convention, the linear space of mass-conserving generators is

$$\mathcal{Q}_n = \{Q : \varepsilon Q = 0\}, \quad \dim \mathcal{Q}_n = n(n-1).$$

A basis is $E_{ij} - E_{jj}$ for $i \neq j$. Thus $\dim \mathcal{Q}_3 = 6$. Nonnegative off-diagonal coefficients select the continuous-time Markov-generator cone inside the corresponding real space; an arbitrary complex element is not a Markov generator.

For $A \in \mathcal{A}_\varepsilon$ with column-sum scalar $\lambda(A)$, the map

$$A \mapsto A - \lambda(A)I$$

identifies $\mathcal{A}_\varepsilon/\mathbb{C}I$ with $\mathcal{Q}_n$ as Lie algebras. It preserves brackets because scalar matrices commute and $\lambda([A, B]) = 0$. This explains a six-dimensional reset-compatible sector without claiming that it equals the six-dimensional symmetric-square representation discussed later.

5 Eight traceless endpoint directions

Every $A \in M_3(\mathbb{C})$ has the unique decomposition

$$A = \frac{\text{tr } A}{3}I + \left(A - \frac{\text{tr } A}{3}I\right). \quad (11)$$

Consequently

$$M_3(\mathbb{C}) = \mathbb{C}I \oplus \mathfrak{sl}_3(\mathbb{C}), \quad \dim_{\mathbb{C}} \mathfrak{sl}_3(\mathbb{C}) = 8.$$

An explicit basis is the six off-diagonal units together with

$$H_1 = E_{00} - E_{11}, \quad H_2 = E_{11} - E_{22}.$$

The bracket formula

$$[E_{ij}, E_{k\ell}] = \delta_{jk}E_{i\ell} - \delta_{\ell i}E_{kj} \quad (12)$$

shows how directed transfers generate diagonal contrasts. For example, $[E_{01}, E_{10}] = H_1$.

The traceless projection $\pi(A) = A - (\text{tr } A/3)I$ preserves the Lie bracket:

$$\pi([A, B]) = [\pi(A), \pi(B)].$$

It does *not* preserve arbitrary associative products. Traceless matrices are not closed under multiplication, as H_1^2 already demonstrates. Thus “subtracting one from nine” has an exact Lie-algebra interpretation, not a claim that one may discard a dimension while retaining every multiplication rule unchanged.

For an optional real observable convention, the Hermitian traceless space has the basis

$$\begin{aligned} X_{ij} &= E_{ij} + E_{ji}, & Y_{ij} &= -i(E_{ij} - E_{ji}) \quad (i < j), \\ D_1 &= E_{00} - E_{11}, & D_2 &= \frac{E_{00} + E_{11} - 2E_{22}}{\sqrt{3}}. \end{aligned}$$

It has eight real dimensions and is closed under $-i[A, B]$. Multiplication by i identifies it with the real Lie algebra $\mathfrak{su}(3)$. This is a mathematical choice of real form. The derivation does not show that an implementation is unitary, that its available controls generate all physical $SU(3)$ operations, or that an eight-dimensional quantity has been observed in nature.

Interpretation. The exact endpoint claim is $9 = 1 + 8$. Nine counts all linear maps; eight counts the non-scalar directions under the trace decomposition. A physical interpretation requires extra dynamics, observables, units, and empirical evidence.

6 Retaining cycles: from endpoint matrices to a path algebra

6.1 Why the finite matrix model is not a faithful history model

In the endpoint representation,

$$E_{02}E_{21}E_{10} = E_{00}.$$

A completed cycle beginning at b_0 therefore has the same image as the length-zero identity path at b_0 . A phase return has been identified with no phase movement at all. This is acceptable for endpoint reasoning but unacceptable as a complete representation of the path taken.

Let Q be the directed three-cycle in (2). Its path algebra $\mathbb{C}Q$ is the vector space with all finite directed paths as a basis, including a length-zero idempotent at each vertex. Multiplication concatenates composable paths and is zero otherwise, using right-to-left operator composition. This is the standard path-algebra construction [4, 8].

6.2 An exact polynomial representation

Introduce a central formal variable z . Give the designated crossing $b_2 \rightarrow b_0$ a factor of z :

$$a_0 \mapsto E_{10}, \quad a_1 \mapsto E_{21}, \quad a_2 \mapsto zE_{02}, \quad v_j \mapsto E_{jj}. \quad (13)$$

Here v_j is the length-zero path at vertex j . The aggregate weighted shift is

$$T(z) = \begin{pmatrix} 0 & 0 & z \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad T(z)^3 = zI. \quad (14)$$

The variable counts crossings of the chosen cut. On a closed path it counts full cycles. On an open path its value depends on the chosen starting vertex and cut; it should not be described as a coordinate-free winding number without that qualification. It is also not automatically the existing ZBA verification-depth field, which has different update semantics.

Theorem 6.1 (Faithful cycle representation). *The map (13) embeds $\mathbb{C}Q$ into $M_3(\mathbb{C}[z])$. Its image is*

$$\mathcal{A}_{\text{cyc}} = \{A(z) : A(0) \text{ is lower triangular}\} = \text{LT}_3(\mathbb{C}) + zM_3(\mathbb{C})[z]. \quad (15)$$

A path starting at j , of length $L \geq 0$, has image

$$z^k E_{ij}, \quad i = (j + L) \bmod 3, \quad k = \left\lfloor \frac{j + L}{3} \right\rfloor. \quad (16)$$

Proof. Each arrow contributes its matrix unit and contributes one factor of z precisely on crossing $2 \rightarrow 0$. Repeated use of (3) gives (16). From the image one recovers the source j , target i , and length

$$L = 3k + i - j.$$

The directed simple cycle has exactly one path for each admissible source and length. Distinct paths therefore map to distinct polynomial matrix monomials, which are linearly independent. This proves injectivity on the path algebra, not merely on its individual paths.

If $i \geq j$, a path to i from j exists without crossing the cut, and every additional full cycle increases k by one. If $i < j$, at least one crossing is required. Hence the admissible monomials are exactly $z^k E_{ij}$ with $k \geq 0$ below or on the diagonal and $k \geq 1$ above it. These span precisely (15). $\square$

Proposition 6.2 (Center and endpoint specialization). *The center of $\mathcal{A}_{\text{cyc}}$ is $\mathbb{C}[z]I$. Evaluation at $z = 1$ is a surjective algebra map with kernel $(z - 1)\mathcal{A}_{\text{cyc}}$. Consequently*

$$\mathcal{A}_{\text{cyc}}/(z - 1)\mathcal{A}_{\text{cyc}} \cong M_3(\mathbb{C}). \quad (17)$$

Proof. A central element must commute with the three vertex idempotents, so it is diagonal. Commutation with E_{10}, E_{21}, zE_{02} forces all three diagonal polynomial entries to be equal. Conversely, scalar polynomial matrices are central.

Evaluation at one reaches every matrix unit, since zE_{ij} evaluates to E_{ij} for the upper-triangular units. A matrix polynomial vanishing at one is entrywise divisible by $z - 1$. Above the diagonal, an entry has the form $zf(z)$. Since z and $z - 1$ are relatively prime, the quotient by $z - 1$ still has a factor of z . The quotient therefore remains in $\mathcal{A}_{\text{cyc}}$, proving the kernel statement. $\square$

Equation (17) locates the exact information loss: setting the cycle variable to one declares each full cycle indistinguishable from its source identity. The eight-dimensional endpoint sector is obtained only after this deliberate identification.

6.3 The trace-zero history algebra is polynomial Iwahori

Define

$$\mathfrak{g}_{\text{cyc}} = \{A(z) \in \mathcal{A}_{\text{cyc}} : \text{tr } A(z) = 0\}.$$

Let $\mathfrak{b}_- \subset \mathfrak{sl}_3(\mathbb{C})$ denote the lower-triangular traceless matrices. Then

$$\boxed{\mathfrak{g}_{\text{cyc}} = \mathfrak{b}_- + z\mathfrak{sl}_3(\mathbb{C})[z]}. \quad (18)$$

This is the lower-triangular version of the standard polynomial Iwahori construction: the constant term lies in a Borel subalgebra [6, 7]. The identification is with an existing algebra, not a claim to invent that algebra.

As a $\mathbb{C}[z]$ -module, $\mathfrak{g}_{\text{cyc}}$ is free of rank eight, with basis

$$H_1, H_2, E_{10}, E_{20}, E_{21}, zE_{01}, zE_{02}, zE_{12}. \quad (19)$$

As a complex vector space it is infinite-dimensional. Its ordinary polynomial-degree Hilbert series is

$$H_{\mathfrak{g}_{\text{cyc}}}(t) = 5 + \frac{8t}{1-t} = \frac{5+3t}{1-t}. \quad (20)$$

The constant grade has five dimensions; every strictly positive grade has eight. Thus it would be incorrect to say that the positive-cycle algebra has eight dimensions at *every* grade. It has eight module generators, three of which begin at degree one.

Evaluation at $z = 1$ induces

$$\mathfrak{g}_{\text{cyc}}/(z - 1)\mathfrak{g}_{\text{cyc}} \cong \mathfrak{sl}_3(\mathbb{C}).$$

The proof follows the same entrywise divisibility argument as Proposition 6.2; dividing a zero trace by $z - 1$ still gives zero trace.

6.4 Formal cycle inversion and the loop algebra

Adjoin z^{-1} . The upper-triangular units are then obtained from zE_{ij} by multiplication by z^{-1} , so

$$\mathcal{A}_{\text{cyc}}[z^{-1}] = M_3(\mathbb{C}[z, z^{-1}]), \quad (21)$$

$$\mathfrak{g}_{\text{cyc}}[z^{-1}] = \mathfrak{sl}_3(\mathbb{C}) \otimes_{\mathbb{C}} \mathbb{C}[z, z^{-1}]. \quad (22)$$

The latter is the loop algebra of $\mathfrak{sl}_3$, with

$$[Xz^m, Yz^n] = [X, Y]z^{m+n}.$$

It has eight directions at every Laurent degree $m \in \mathbb{Z}$. No central extension or affine Kac–Moody derivation is assumed here. Negative powers are a formal algebraic enlargement, not permission to delete audit events or execute a forbidden reverse transition.

A compact generating-function identity follows from (14):

$$(I - uT(z))^{-1} = \frac{I + uT(z) + u^2T(z)^2}{1 - zu^3}. \quad (23)$$

This is valid as a formal power series in u with polynomial matrix coefficients. It enumerates endpoint changes and repeated cut crossings. It is not, without an additional model, a physical propagator or measured frequency law.

6.5 A limitation and a faithful labeled extension

The polynomial construction faithfully retains *phase paths on the simple cycle*. It does not retain every ZBA event. For example, two different authorized operations with the same phase arrow have identical images if both are represented by E_{10} . Their policies, payloads, and actors are not reconstructed from z .

For a finite directed graph with separately labeled edges a , assign a distinct noncommuting variable x_a to each edge. Let $\mathbb{C}\langle x_a \rangle$ be the free associative algebra on those variables. Map an edge from j to i to

$$E_{ij}x_a \in M_n(\mathbb{C}\langle x_a \rangle).$$

A path a_1 then a_2 through a_L maps to

$$E_{ij}x_{a_L} \cdots x_{a_1},$$

where the reversed written order follows right-to-left operator composition. Distinct edge paths yield distinct words together with their endpoints; vertex identities remain distinct. Hence this is a faithful representation of that graph's path algebra. This standard free-word construction supplies a direct route from the one-variable cycle model to an event-labeled model. It does not authenticate the event record or prove that an asserted external event actually occurred.

7 Reset, polarity, and the missing policy hypothesis

The ZBA reference reset chooses checked or rejected evidence while setting polarity and phase to their neutral boundary values [3]. Once arbitrary policies are admitted, commutation requires a condition on that choice.

Let $v : S \rightarrow \{0, 1\}$ be a deterministic validation predicate, and let $\sigma(1) = \text{checked}$, $\sigma(0) = \text{rejected}$. Define

$$R_v(p, b, d, e) = (0, b_0, d, \sigma(v(p, b, d, e))).$$

Assume both composites discussed below are defined; for genuinely partial reset domains, equality of the applicable domains is an additional requirement.

Theorem 7.1 (Policy equivariance and reset stability). *For the total reduced-state model,*

$$R_v M = M R_v \iff v(Ms) = v(s) \text{ for every } s, \quad (24)$$

$$R_v^2 = R_v \iff v(R_v s) = v(s) \text{ for every } s. \quad (25)$$

Proof. The state $MR_v(s)$ equals $R_v(s)$ because its polarity is already zero. All components of $R_vM(s)$ and $MR_v(s)$ agree except potentially their evidence, which is respectively $\sigma(v(Ms))$ and $\sigma(v(s))$. Since σ is injective, equality is equivalent to (24). The same comparison between $R_v^2(s)$ and $R_v(s)$ proves (25). $\square$

7.1 A stable reset that does not commute with the mirror

Consider

$$v(s) = \mathbf{1}\{e \neq \text{rejected}\} \mathbf{1}\{p \neq -1\}. \quad (26)$$

After successful reset, polarity is zero and evidence is checked, so validation stays successful. After unsuccessful reset, evidence is rejected, so validation stays unsuccessful. Thus the policy is reset-stable and $R_v^2 = R_v$.

For $s = (+1, b_1, 0, \text{claimed})$, however,

$$R_vM(s) = (0, b_0, 0, \text{rejected}), \quad MR_v(s) = (0, b_0, 0, \text{checked}).$$

Therefore $R_vM \neq MR_v$.

This strengthens the specification rather than contradicting its actual reference implementation. The inspected reference rule depends only on whether evidence is already rejected, so it is mirror-invariant. Its commutation property remains valid. The correction concerns any broader formulation in which “validation succeeds” is left policy-dependent without requiring mirror invariance. Stable validation alone is insufficient.

A policy profile can now declare either *mirror equivariance*, tested by $v \circ M = v$, or *order sensitivity*, in which the two composites may intentionally differ and their order must be retained. For example, a policy that treats the two polarities as different authority classes should not be forced into a false commutation axiom.

8 Why structural reversibility cannot erase an append-only audit

Assume a mirror-invariant, reset-stable structural model so that

$$M^2 = I, \quad R^2 = R, \quad MR = RM = R. \quad (27)$$

The generated transformation monoid has exactly three elements $\{I, M, R\}$ provided these transformations are distinct. In particular, it is not the six-element triangle group.

Theorem 8.1 (Exponential history collapse). *Among the 2^L operation words of length L over $\{M, R\}$, exactly $2^L - 1$ contain a reset and evaluate to the same transformation R . The remaining word is M^L .*

Proof. There is exactly one word without R , namely the all- M word. In every other word, the absorbing relations $MR = RM = R$ and $R^2 = R$ reduce the entire product to R . The all- M word reduces to I for even L and M for odd L . $\square$

At $L = 12$, one structural result represents 4095 distinct operation histories. For a uniformly selected history in this reset fiber, the conditional uncertainty is

$$\log_2(2^L - 1) \text{ bits.}$$

Any fixed-length, injective binary identifier for these histories requires at least $\lceil \log_2(2^L - 1) \rceil = L$ bits when $L \geq 2$. This is a combinatorial distinguishability bound. It does not assert that a cryptographic hash can reconstruct a history, that hash collisions are impossible, or that a finite-dimensional real-valued vector space can never encode an infinite set.

Theorem 8.2 (Append-only obstruction). *Let an audited lift of each allowed operation append a nonempty record to a history. Whenever the relevant composites are defined, an audited mirror cannot be an exact involution on the full state-plus-history space, and an audited reset cannot be exactly idempotent there. If mirror and reset append distinct fixed operation labels, their two execution orders also have different histories.*

Proof. Write an audited update as $\hat{f}(s, h) = (f(s), h \parallel \sigma_f)$ with σ_f nonempty. Two applications append more history than zero applications, so $\widehat{M}^2$ cannot be the full identity. Two reset applications append more history than one, so $\widehat{R}^2 \neq \widehat{R}$. Finally the suffixes $\sigma_M \sigma_R$ and $\sigma_R \sigma_M$ differ for distinct single-operation labels, even when their structural states agree. $\square$

The obstruction is a consequence of an append-only requirement, not a theorem that reversible computation is impossible. Systems can reject redundant operations, record an explicit no-op, or define an equivalence that deliberately ignores permitted audit extensions. Those choices must be stated. Deleting the history would remove the very invariant the audited model is intended to preserve.

The phase-cycle result and the mirror/reset result illuminate different information losses. A pure cycle requires source, endpoint, and crossing count to reconstruct its phase path. Branching histories over multiple operations can carry far more information. Neither a single phase tag nor the one-variable polynomial model constitutes a complete provenance record for an arbitrary workflow.

9 Minimal phase distinctions and the limits of numerical encoding

Suppose each of the three phases has a different next enabled action: Emerge at b_0 , Enter at b_1 , and Seal at b_2 , under the successful-cycle assumptions. Let $q : B \rightarrow X$ be an encoding from which the enabled action must be determined, with no additional phase information available.

Proposition 9.1 (Three behavioral classes are necessary). *Such an encoding must be injective, and therefore $|X| \geq 3$.*

Proof. If $q(b_i) = q(b_j)$ for distinct phases, a deterministic decision function on X must return the same enabled-action set for both. The required sets differ, giving a contradiction. $\square$

This provides an operational justification for preserving three tags even though scalar erasure maps all of them to zero. It does not privilege the typography $-0, 0, +0$ over the ordinary strings entering, boundary, emerging. An ordinary typed three-state machine can preserve exactly the same information.

Similarly, retaining z formally is different from evaluating it at a number. Setting $z = 1$ erases all crossing counts. Setting it to a root of unity of order m identifies counts differing by m . In a pure cycle, storing the integer count explicitly needs only logarithmically many bits in its maximum value; retaining an arbitrary word of independently chosen event labels can require linearly many bits in word length. These are different compression problems.

10 Where the separate $SU(3)$ sequence does and does not fit

For comparison, the finite-dimensional irreducible complex representations of $SU(3)$ are labeled by nonnegative integers (p, q) , with dimension

$$d(p, q) = \frac{(p+1)(q+1)(p+q+2)}{2} \quad (28)$$

[9]. This gives the first distinct nontrivial dimensions 3, 6, 8. There is no nine-dimensional irreducible representation: a label at least three gives dimension at least ten, and checking the remaining pairs yields only 1, 3, 6, 8 below ten.

For the defining representation $V = \mathbb{C}^3$,

$$\dim V = 3, \quad \dim \text{Sym}^2 V = 6, \quad \dim \text{End}_0 V = 8.$$

The usual decompositions are

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}, \quad \mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{1} \oplus \mathbf{8}.$$

These established representation-theoretic facts are consistent with the endpoint trace split. They do not prove that ZBA is a fundamental physical theory or that its six-dimensional reset sector transforms as $\text{Sym}^2 V$.

There is an instructive exact coincidence. The counts of symmetric pairs and ordered distinct pairs in dimension n are respectively

$$\frac{n(n+1)}{2} \quad \text{and} \quad n(n-1).$$

For positive n , equality holds precisely at $n = 3$. Both then equal six. Nevertheless, under relabelling by S_3 the two six-dimensional spaces are not isomorphic. On the identity, a transposition, and a three-cycle, their characters are

$$\chi_{\text{Sym}^2}(1, \tau, \rho) = (6, 2, 0), \quad \chi_{\text{directed}}(1, \tau, \rho) = (6, 0, 0).$$

A transposition fixes one diagonal symmetric pair and one off-diagonal unordered pair, but fixes no ordered pair of distinct vertices. The unequal traces prove inequivalence. This is a concrete test against treating every appearance of six as the same structure.

11 Independent exact verification

Two standard-library Python programs accompany this article. They use exact integers, rational row reduction, explicit path enumeration, and polynomial matrices. The first independently implements the displayed structural state rules; it is not a byte-verified execution of the upstream files. The second tests the history-algebra extension. Both completed successfully in Python 3.13.5 during preparation.

Check	Observed result
Independent reference-model laws	144 states; 432 successful instances of $M^2 = I$, $R^2 = R$, and $MR = RM$.

Check	Observed result
Mirror-sensitive policy (26)	Reset idempotence passed for all 144 states; commutation passed for 72 and failed for 72. The equivariance equivalence held in every case.
Permutation/reset/guard dimensions	(2, 3, 4) for $n = 2$; (5, 7, 9) for $n = 3$; (10, 13, 16) for $n = 4$; (17, 21, 25) for $n = 5$.
Boundary-filter matrix units	203 identities checked across $n = 2, \dots, 8$.
Endpoint Lie algebra	Rank eight; 64 pairwise basis-bracket closure checks.
Three-state generator space	Rank six, with zero column sums.
Simple-cycle paths	39 paths of length at most 12; all images distinct before specialization.
Path multiplication	507 composable and 1014 noncomposable ordered products checked.
Polynomial history algebra	Eight module generators; 64 bracket checks; homogeneous dimensions (5, 8, 8, 8, 8) at degrees zero through four.
Labeled extension	48 paths of length at most six on a graph with parallel labeled edges; all free-word matrix images distinct.
Mirror/reset words	8191 words through length 12; structural monoid size three and reset fibers $2^L - 1$.
$SU(3)$ dimension formula	441 label pairs checked; first distinct dimensions 1, 3, 6, 8, 10. The universal exclusion of nine follows from the proof, not this finite search alone.

The exact counts document what was run. They are not measurements on a deployed service, a benchmark against other tools, or a cryptographic audit. The original repository’s reported six-record positive chain and 14 mutated-record cases were read but were not rerun here. No conclusion about all possible policy predicates follows from enumerating one policy’s finite examples; the universal statement is instead proved in Theorem 7.1.

Reproduction from the package root requires only:

```
python code/verify_368.py --output results/verification.json
python code/verify_history_algebra.py \
  --output results/history_algebra.json
```

The programs use Python assertions; they must be run without the optimization option `-O`, which disables assertions. JSON outputs include the SHA-256 of each script. The package also contains a source ledger, claim-status table, and checksums of the delivered files.

12 Engineering value and a falsifiable research programme

12.1 A specification improvement available immediately

The policy theorem supplies a concrete conformance obligation. A profile claiming mirror/reset commutation must establish mirror-invariant validation and compatible domains. A profile

claiming reset idempotence must establish reset-stable validation. These should be separate declarations and separate tests: the counterexample proves one does not imply the other.

A useful compiler pass could construct the phase-only graph, enumerate guarded transitions, and calculate its generated endpoint algebra. The pass would report which domain distinctions were erased by replacing partial operations with total updates. For the three-cycle, a total permutation-only representation cannot substitute for the guarded representation without losing this distinction. The exact matrix-unit formula gives a small verification certificate.

12.2 A history abstraction with an explicit contract

A phase-path abstraction can store (j, i, k) , where j is the source, i the target, and k the designated crossing count. On the simple cycle this is lossless, by Theorem 6.1. On a graph with parallel event labels it is not. A tool should therefore declare whether it retains endpoints, simple-cycle paths, event-label words, or full authenticated records. These are different contracts, not successive marketing descriptions of one object.

The algebraic layer can sit beside established provenance interchange. W3C PROV distinguishes entities, activities, and agents [10]; the proposed operators organize phase and policy reasoning without replacing that provenance vocabulary. Integrity, authorization, and trustworthy external evidence remain separate engineering requirements.

12.3 A controlled comparison rather than a symbolic advantage claim

A deployment study should compare an ordinary typed finite-state machine with a ZBA-labeled implementation *holding the transition graph, policy checks, log fields, and cryptographic mechanisms constant*. This prevents a richer implementation from being credited to its numeral labels. A third condition may add the proposed algebraic compiler and abstraction checks to test the actual new tooling.

The experimental unit should be a complete workflow or an independently written policy, not every correlated row from the same generated chain. Synthetic violations should cover phase misordering, parallel-label aliasing, mirror-sensitive policies, reset-unstable policies, and improper reuse of endpoint equality as history equality. If the evaluation concerns authentication, attackers must also be allowed to recompute unkeyed hashes; otherwise changing a field while preserving its old digest tests only a narrow integrity condition.

Outcomes should include missed violations, false rejections, diagnosis accuracy, time to identify a seeded fault, verifier runtime, memory, and record-size overhead. Functional equivalence of matched baselines should be checked before comparing human diagnosis. An empirical benefit is supported only by an observed difference attributable to the added tool, not by the existence of an isomorphic mathematical description.

A concrete falsifiable hypothesis is: *explicit abstraction labels and policy-equivariance certificates reduce errors caused by conflating structural equality with audited equality*. The null is no improvement over an ordinary state-machine interface exposing the same information. No such user study or deployment comparison has been conducted in this article.

13 Novelty, limitations, and precise claim boundaries

The inspected ZBA 1.1 paper and reference code did not supply the particular completion ladder, polynomial cycle representation, Iwahori identification, policy-equivalence theorem, and executable tests assembled here. This supports describing them as proposed additions to that

inspected specification. It does not establish priority over all earlier literature, private work, or every file in the author’s broader research corpus.

The literature search checked the immediate mathematical antecedents: representations and path algebras, matrix-algebra generation, polynomial Iwahori algebras, loop/path constructions, and $SU(3)$ dimensions. It was a targeted comparison, not an exhaustive bibliographic or patent novelty search. The already-known names and structures are cited explicitly. A strongest-form historical claim such as “the first discovery of an eight-dimensional boundary algebra” is not supported.

Several model restrictions matter. The complex linear span is richer than the set of valid state operations. The successful-cycle model omits genesis, recovery, and rejection branches. The single variable records phase crossings but not arbitrary event labels. The optional boundary reflection is not the existing polarity mirror. The existing recursion depth is not silently repurposed as a crossing counter. An exact history word is not a collision-resistant commitment, and a valid commitment is not proof of the truth of an external event.

No physical measurements appear here. Eight endpoint directions do not establish eight fundamental forces, a new cosmology, a quantum effect, or a replacement for ordinary arithmetic. No encryption primitive or security amplification follows from the numbers three, six, or eight. These exclusions are consequences of what the model contains: it supplies algebraic operations and proofs, not dynamical laws or experimental observables.

The constructive novelty claim that remains is substantial but specific: *a theorem-led ZBA extension that separates guarded-state completion from history retention, identifies the precise endpoint quotient, and makes policy-dependent algebraic laws testable*. Its immediate scientific value is explanatory and falsifiable: it can show exactly where a representation forgets information and exactly when a proposed identity fails.

14 Conclusion

The strongest mathematical interpretation of “3, 6, 8” is not an alternative multiplication table. Three typed phases support a guarded transition algebra. Total resets preserve a common-column-sum constraint; source-sensitive guards remove it. In dimension three, the resulting non-scalar endpoint space has eight directions, while the corresponding total-reset sector has six. Separately, the familiar $SU(3)$ irreducible dimensions provide a consistent but logically distinct appearance of the same triple.

The deeper result concerns history. The full directed-cycle path algebra is faithfully represented by polynomial matrices with triangular constant term. Its trace-zero part is polynomial Iwahori; specializing the crossing variable to one produces the eight-dimensional endpoint algebra. Thus eight is a precise quotient-level answer, not the size of all possible auditable histories.

Finally, reset commutation is a property of the validation policy as well as the coordinate assignments. The necessary-and-sufficient equivariance condition prevents a false generalization, and the exponential reset fiber quantifies how much history can disappear behind one structural state. Together these results turn a numerical intuition into a concrete research programme: define the operators, expose the quotient, prove the invariants, and test the lost distinctions.

A return to the boundary is not a return to the past.

A Notation and conventions

Symbol	Meaning
P, B	Polarity labels and three boundary-phase tags; separate state factors.
M, J	Existing polarity mirror and optional boundary reflection; not the same operation.
C, C_B	Cyclic shift matrix and its phase-state counterpart.
R, R_v	Structural reset under the reference or specified validation rule.
E_{ij}	Matrix unit mapping source basis vector e_j to target e_i .
P_0	Boundary-domain filter E_{00} , not a constant reset.
K_i	Constant reset matrix $e_i \varepsilon$.
$\mathcal{P}_n, \mathcal{A}_\varepsilon$	Permutation span and common-column-sum algebra.
π	Traceless projection; Lie-linear but not an associative homomorphism.
z	Formal count of crossings of the edge $2 \rightarrow 0$; not a physical constant or automatically the verification depth.
$\mathcal{A}_{\text{cyc}}$	Faithful polynomial representation of the simple directed-cycle path algebra.
$\mathfrak{g}_{\text{cyc}}$	Its trace-zero polynomial Iwahori subalgebra.
$\hat{f}$	Audited execution that retains structural action and appended history.

Products act right to left. Dimensions are complex vector-space dimensions unless explicitly described as real or as module rank. The variable z commutes with coefficients; the variables x_a in the fully labeled extension do not commute. “Endpoint” refers to this phase representation, not to every component of a complete ZBA record.

B Source scope and reproducibility record

The upstream specification inspected was the GitHub file in the `papers/` directory named

`zero-boundary-algebra-formal-specification-1.1.md`.

It was labeled version 1.1.0 and dated 18 August 2026. Its text, the Hugging Face dataset card, the displayed reference program, and the displayed property-check program were accessed through web retrieval on 27 September 2026. The Hugging Face resolver exposed revision

`12ac9a62d2cfbf50f3d33fbc6db19b406d31c4b1`.

No exact GitHub commit was verified. Original artifact bytes were not available to the execution environment; the package does not claim to contain an upstream checkout or byte-identical rerun. The tests supplied here are new independent implementations and are labeled accordingly. PDF pages containing the original state sorts and reset/mirror propositions were also visually inspected during source review.

The package separates the article (PDF, LaTeX, and Markdown), two verification programs, JSON results, and a source ledger into the `article/`, `code/`, `results/`, and `sources/` directories. The root contains a reproduction guide, a claim-status table, and `SHA256SUMS.txt`. The source ledger distinguishes retrieved text from independently executed code.

The editable LaTeX is the typesetting source of record. The Markdown version is a convenience export; numbered equations, theorem counters, and cross-references are authoritative in the PDF

and LaTeX. AI-assisted preparation included source retrieval, derivation, coding, and editorial synthesis. This disclosure does not substitute for the proofs or the executable checks.

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